

DAT 5564: Database Design and SQL

Fall 2025, Mini A

Homework#4: SQL Part 2

Instructions

1. This is an individual assignment. You may not discuss your approach to solving these questions with anyone (or Generative AI), other than the instructor or TA.
2. Please include only your Student ID in the submission.
3. **The only allowed material is:**
 - a. Class notes
 - b. Content posted on the LMS
 - c. Textbook
4. You are not permitted to use other online resources.
5. **There will be TA office hours. See the course schedule.**

Background

The E-commerce Order Dataset provides structured information regarding orders, customers, products, payments, and order fulfillment for an online retail platform. The dataset is designed to facilitate comprehensive analysis of e-commerce operations, tracking customer behavior, order processing, and logistics.

This schema enables comprehensive analysis of customer purchasing behavior, seller performance, order fulfillment efficiency, and payment trends. Researchers and analysts can use this dataset to derive insights into e-commerce operations, optimizing business strategies and improving customer satisfaction.

Dataset Structure: The dataset consists of multiple relational tables, each focusing on different aspects of e-commerce transactions.

Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date)

This table records details related to customer purchases and order fulfillment.

- **order_id**: Unique identifier for each order.
- **customer_id**: Unique identifier for the customer placing the order.
- **order_status** : Status of the order (e.g., delivered, cancelled, processing).
- **order_purchase_timestamp** : Timestamp when the order was placed.

- **order_approved_at** : Timestamp when the order was approved.
- **order_delivered_timestamp** : Timestamp when the order was delivered.
- **order_estimated_delivery_date** : Expected delivery date communicated to the customer.

OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)

Captures individual items within an order, linking products to transactions.

- **order_id**: Identifier linking items to an order.
- **product_id**: Unique identifier for the purchased product.
- **seller_id** : Unique identifier for the seller of the item.
- **price** : Price of the product.
- **shipping_charges** : Shipping cost associated with the item.

Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state)

Stores information about customers registered on the platform.

- **customer_id** : Unique identifier for each customer.
- **customer_zip_code_prefix** : Zip code prefix of the customer.
- **customer_city** : City where the customer resides.
- **customer_state** : State where the customer resides.

Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)

Contains details of products available for sale.

- **product_id** : Unique identifier for each product.
- **product_category_name** : Name of the product category.
- **product_weight_g** : Weight of the product in grams.
- **product_length_cm** : Length of the product in centimeters.
- **product_height_cm** : Height of the product in centimeters.
- **product_width_cm** : Width of the product in centimeters.

Relations:

The dataset follows a relational structure, ensuring referential integrity among tables:

Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_timestamp, order_estimated_delivery_date)

OrderItems (order_id, order_item_id, product_id, seller_id, price, shipping_charges)

Customers (customer_id, customer_zip_code_prefix, customer_city, customer_state)

Products (product_id, product_category_name, product_weight_g, product_length_cm, product_height_cm, product_width_cm)

Database: The E/R Diagram for the database is below.



In Homework Part #1, submit answers to these questions online on Canvas by the due date. They are automatically graded.

In Homework part #2, include only answers to questions 7 and beyond in a PDF file and submit on the Part #2 page on Canvas.

*For each question, submit your SQL code and a screenshot of the results. If the results are too long, partial results are fine. Include relevant attributes for each result to explain that the result is correct. Do NOT include many unnecessary attributes. Do NOT use SELECT *.*

Assignment

- 1- How many distinct orders were placed from January 1, 2017, to December 31, 2017, that were delivered and had shipping charges above 60?
- 2- List product_id and category of products where the category name contains the word “bed” and weighs more than 1,000 grams. Sort by the order_id.
- 3- Find the total number of customers in each state. List only states with more than 10 customers.
- 4- During 2018, what was the largest total cost (including shipping charge) of items in a single order? Your query should return only one number.
- 5- List a pair of customers living in the same state, but in different cities. Order by the customer's state. List customer ID, customer city, and customer state.
- 6- Find out how many late orders and how many early orders are on the orders table. We call an order early if the delivery happens before the estimated time; otherwise, we call it late. Your query should return only two rows and two columns.
- 7- Find the total order costs (price + shipping_charges) by city. List the city and the total order cost. List only cities with order costs of 50 or less.
- 8- Find product categories that have more than the average number of products across all categories. Hint: You need to use a subquery.
- 9- List customer cities where the average price of orders from their citizens exceeds 5 times the average price of all orders. Hint: Only consider price.
- 10- List the top 5 customers who placed the most orders (in terms of total price) among all orders, calculate their total number of orders and total amount spent, and sort them in descending order by total amount they have spent. Only consider price as order cost.

- 11- Find the active sellers who have at least sold 5 products.
- 12- Find the average shipping charge and number of orders per customer city. List only those who had more than 2 orders. Sort by average shipping charges in decreasing order.
- 13- List the products that are heavier than the average weight of all products.
- 14- How much time did it take for you to finish?